

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

16 September 2013

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
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Total	

This document consists of **17** printed pages and **1** blank page.

[Turn over

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Answer **all** questions.

- 1 Plasmids used in genetic engineering as a cloning vector include pMOD2-Neo shown in Fig. 1.1. This plasmid contains genes coding for antibiotic resistance – ampicillin resistance gene (amp) and neomycin resistance gene (neo).

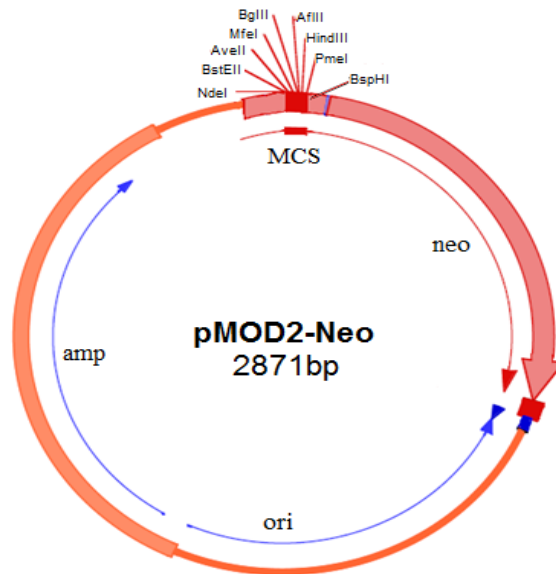


Fig. 1.1

With reference to the plasmid shown in **Fig 1.1**,

(a) State two properties of the plasmid which allows it to be used as a cloning vector. [2]

1. Has an origin of replication;
2. Allows pMOD2-Neo/plasmid and foreign gene/GOI to replicate independently of bacterial genome within the host cell;
3. To give multiple copies of the plasmid and gene within one bacterium;

OR

4. Has an ampicillin resistance gene
5. Allows for the selection of (successfully) transformed bacteria containing (both recombinant and non-recombinant) plasmids;;

R: Selection of transformed bacteria with recombinant plasmid, survival of bacteria not plasmids

OR

6. Has a neo gene
7. Allows for the selection of (transformed) cells with recombinant plasmids;;

OR

8. Has multiple cloning site/restriction sites;
9. can be cut by specific restriction enzymes to insert foreign gene/GOI;

OR

10. has genetic markers
11. Allows for the selection of transformed cells with both recombinant and non - recombinant) plasmids;;

(b) The plasmid pMOD2-Neo and the human DNA containing the gene of interest are cut with restriction enzymes and cloned using bacteria in genetic engineering.

(i) Explain what is meant by a *restriction enzyme*. [2]

- 1. Restriction enzyme recognises and binds;**
- 2. to short sequences of bases of DNA (restriction sites);**
- 3. The restriction site is 4-6 base pairs in length;**
- 4. Catalyses the breaking of phosphodiester bonds between specific nucleotides / cut DNA at specific points;**
- 5. To produce restriction fragments with blunt/sticky ends;**

(ii) Suggest why the restriction enzyme will cut the human DNA in many places but will only cut the plasmid DNA once. [1]

- 1. restriction enzymes only cut DNA at specific base sequence/recognition site;**
- 2. sequence of bases/recognition site (on which enzyme acts) occurs once in plasmid but many times in human DNA;**

(iii) Suggest why the plasmid should only be cut once. [1]

- 1. To prevent non-specific and random DNA fragments from being generated;;**
 - 2. To transfer the gene of interest into a specific location in the plasmid;;**
- OWTTE**

- (c) Disease X is an autosomal disease resulting from a single base substitution which removes a recognition site of the restriction enzyme, *Hae*III as shown in Fig. 1.2. The enzyme's recognition sites on allele 1 and allele 2 are indicated by arrows.

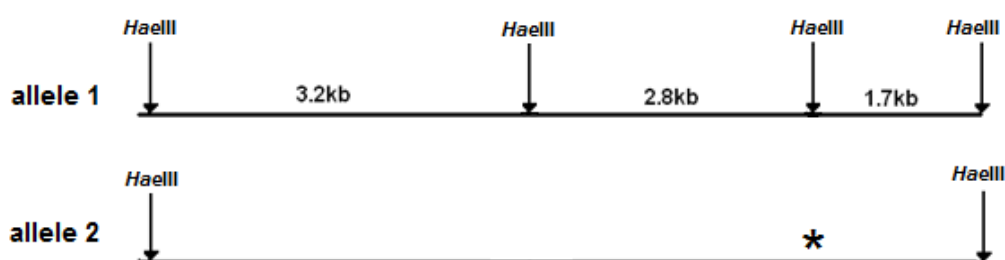


Fig 1.2

The mutation can be detected by restriction fragment length polymorphism after cleaving with *Hae*III. Fig 1.3 shows the banding patterns produced by four individuals.

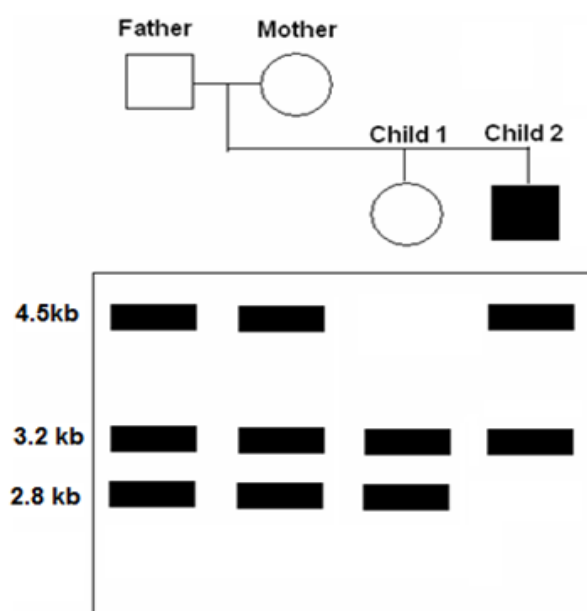
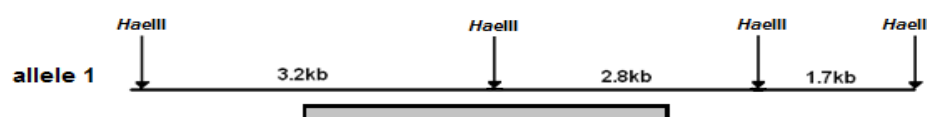


Fig. 1.3

With reference to Fig 1.2 and Fig. 1.3,

- (i) On allele 1, draw the probe at the position it will hybridise to. [1]



- (ii) On allele 2, draw the position of the polymorphic restriction site by marking it with an asterisk (*). [1]
- (iii) State the mode of inheritance for Disease X. [1]

Autosomal recessive;;

(iv) Explain how gel electrophoresis can be used to distinguish between the two alleles. [4]

1. The two alleles Allele 1 (A1) and Allele 2 (A2), differ in their base sequences such that A1 has 3 *HaeIII* restriction sites while A2 has 2 *HaeIII* restriction sites;;
2. Restriction digestion of the DNA with A1 by *HaeIII* will produce 3 restriction fragments of 3.2kb, 2.8kb and 1.7kb;
3. Restriction digestion of the DNA with A2 by *HaeIII* will produce 2 restriction fragments of 3.2kb and 4.5kb;
4. These fragments are separated out by gel electrophoresis, the 4.5kb fragment from A2 is a longer fragment, faces more resistance from gel matrix, will move slower and be found closest to the well (or vice versa);;
5. Different band patterns for A1 and A2 will result after gel electrophoresis;
6. and when the DNA fragments are stained with ethidium bromide, they can be visualised under UV light;

(d) Information from the Human Genome Project (HGP) has allowed for better understanding of the genetic basis of diseases such as Disease X.

Discuss two other benefits arising from HGP. [2]

1. Ref to Genetic Testing + elaboration;;
2. Ref to Gene Therapy + elaboration;;
3. Ref to Pharmacogenomics+ elaboration;;
4. Ref to DNA forensics + elaboration;;

[Total: 15]

- 2 Haemophilia B is a blood clotting disorder. Haemophilia B patients are unable to produce enough human clotting factor IX, also referred to as FIX, which is an essential protein in the blood clotting process.

The dominant allele coding for FIX was packaged in a modified adeno-associated virus, AAV8, that has been designed to infect liver cells specifically. The modified AAV8 was injected into six patients with severe haemophilia B who were producing FIX at less than 1% of normal levels.

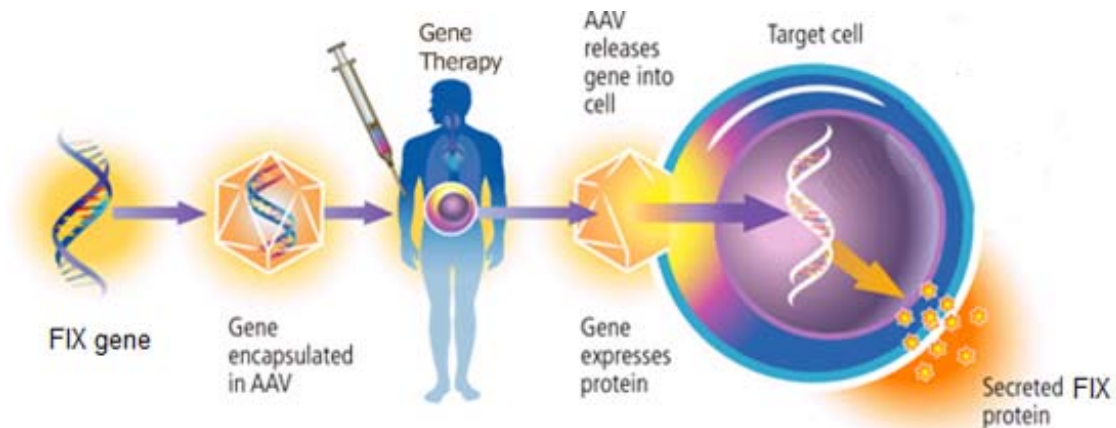


Fig 2.1

Prior to undergoing gene therapy, all six patients were receiving the standard treatment for severe haemophilia B— injection of purified FIX produced by recombinant DNA technology—several times a month. After receiving the FIX gene therapy, each patient produced FIX at between 2% to 12% of normal levels. For at least six months, four of the six patients no longer required purified FIX injections for routine bleeding. The other two patients required purified FIX injections less frequently than before the therapy.

- (a) Suggest and explain why two out of the six patients still require purified FIX injection.
[2]

1. Treatment is not as successful;
2. most liver cells had not taken up the dominant allele coding for FIX/functional FIX allele during viral gene delivery / Only some of liver cells contains dominant allele coding for FIX/functional FIX allele;
3. leading to low expression of the dominant allele coding for FIX/functional FIX allele;;
4. Lower concentration of FIX produced;
OR
5. Gene delivered is not long lasting;
6. because the dominant allele coding for FIX/functional FIX allele delivered remained in the cytoplasm is degraded after some time/earlier;;
7. AAV8 have low integration efficiency;

(b) Explain why it is sufficient to insert a single dominant allele coding for FIX into each liver cell. [2]

- 1. Normal functional allele is dominant/the dominant allele of the gene concerned;**
- 2. a single copy of dominant allele is sufficient/can mask the effect of the recessive/mutant allele ;**
- 3. Expression of the normal allele would result in the synthesis of a normal functional protein product;**
- 4. restoring target cell to normal state/correcting functions/altering the phenotype of the target cell;**

(c) Explain why individuals who have been treated by this method of gene therapy do not pass on the dominant allele coding for FIX to their children. [1]

- 1. Treatment targets the liver/somatic cells and not germ cells/gametes;;**
OR
- 2. gametes of these individuals who have been treated will not contain the dominant allele coding for FIX, so it will not get inherited;;**

(d) State two disadvantages of this type of gene therapy. [2]

- 1. allele may be inserted into tissue other than target with unknown consequences;;**
- 2. Gene therapy is commonly conducted on somatic cells. The natural death of treated cells as well as the rapidly dividing nature of somatic cells makes the gene therapy short-lived hence patients need to be treated on a frequent basis;;**
- 3. Viral vector introduced into the tissue may be recognized as foreign particle and trigger immune system to mount an attack which reduces the effectiveness of gene therapy;;**
- 4. Viral vector may regain virulence and cause an immune response in the patient;;**
- 5. low integration efficiency;;**
(Credit: any other valid data analysis / reverse answers)

(e) Suggest and explain the advantages of using a viral delivery system for the dominant allele coding for FIX. [2]

- 1. Specific targeting, the virus is able to deliver genes to specific target tissues;;**
- 2. Greater efficiency, the virus is able to transfer the gene into the cell more effectively;;**
- 3. The virus is able to introduce the normal DNA / allele into the nucleus;;**

[Total: 9]

3 Investigators often report success in culturing plant tissue using root tips. This technique has many commercial applications which include the production of large numbers of genetically identical plants and the culturing of plant cells in a fermenter with culture medium for the production of secondary products.

(a) (i) Suggest one reason why plant tissue culture was successful using organs such as root tips. [1]

1. meristematic cells which are undifferentiated cells/rapidly dividing cells;;
2. totipotent cell able to divide indefinitely / able to differentiate into all cell types of the new plant;;

(ii) Suggest why it is advantageous that plants produced by plant tissue culture are genetically identical. [1]

1. all possess the desirable features of the stock plants/transgenic plants with improved / new desired traits;;
- OR
2. ensures product uniformity, this reliability is an important sales advantage;;

In many stages of plant tissue culture, strict aseptic techniques need to be used. One of the difficulties of culturing plant tissue is in carrying out surface sterilisation of the explant.

To investigate a suitable method of culturing explants, plant tissue samples were taken at three different times of the year and placed on a medium containing no fungicide or antibiotic. Two different sampling methods were used:

- discs cut from leaves wiped with bleach;
- stem tissues extracted with a sterile hypodermic syringe.

The results of culturing the explants are shown in Fig 3.1.

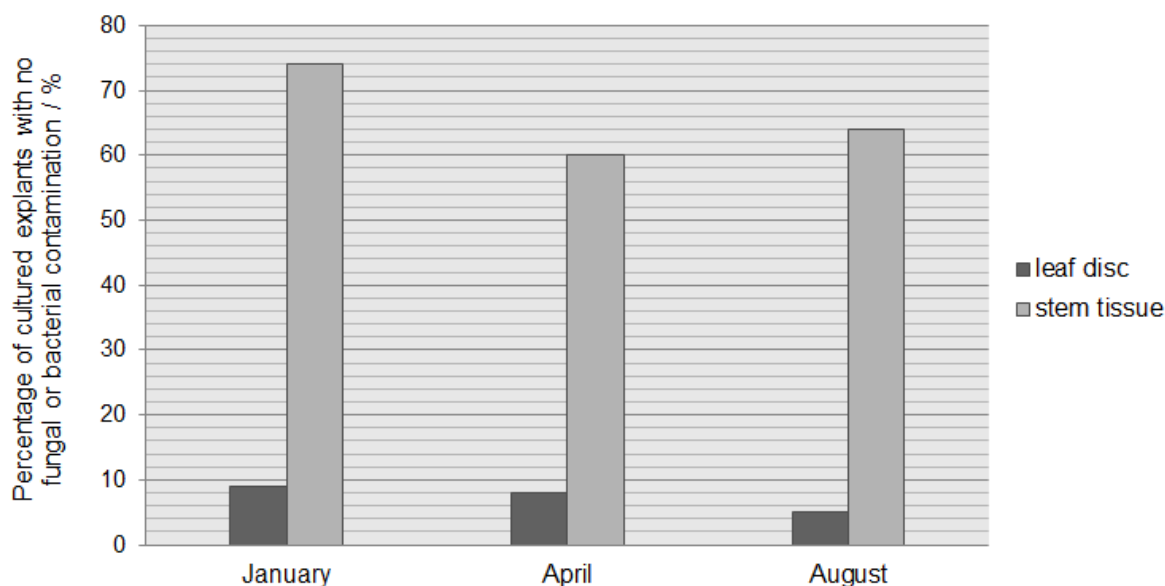


Fig. 3.1

With reference to Fig. 3.1,

- (b) compare the effect of the different methods of taking tissue samples and the time of the year on fungal or bacterial contamination of the cultured explants. [3]

different methods

1. stem tissue has less contamination than leaf discs;
2. average 66.0% vs average 7.33% lesser contaminations / 52% - 65% lesser contaminations;
3. stem tissue has average 9X lesser contamination than leaf discs / 7.5X to 12.8X lesser contamination than leaf discs;
(Credit: any other valid data analysis / reverse answers)
4. extraction with a sterile hypodermic syringe is a better method of taking samples;

time of year

5. both treatments has highest % non-contamination in January;
6. 74% for stem tissue and 9% for leaf disc;
7. lowest % non-contamination for stem tissue is in April while the lowest % non-contamination for leaf discs is in August;
8. 60% for stem tissue and 5% for leaf disc;
(Credit: any other valid data analysis / reverse answers)
9. culturing of explants should be done in January;

- (c) Early attempts to culture plant tissue on a culture medium were not always successful. Studies have shown that the culture medium must contain a range of substances - mineral ions, a sugar suitable for plant cells to use as well as plant growth regulators, to bring about successful plant cell growth and division.

To investigate the role of plant growth regulators, equal masses of plant callus were cultured for four weeks on media containing different concentrations of two plant growth regulators: auxin and cytokinin. The results are shown in Table 3.1.

Table 3.1

treatment	concentration of plant growth regulators / mg dm ⁻³		effect of plant growth regulators on callus growth
	auxin	cytokinin	
A	2.00	0.00	little or no growth
B	2.00	0.02	growth of roots
C	2.00	0.20	increased growth of callus with no differentiation
D	2.00	0.50	growth of shoots
E	0.00	0.20	little or no growth

(i) Suggest why a sugar was included in the culture medium. [1]

1. **Photosynthesis is not yet occurring / occurring at a low rate, therefore sugars is not being synthesised;**
 2. **Sugars are required as energy source for respiration/ATP production/ respiratory substrate;**
- OR**
3. **Source of carbon for carbohydrates like starch, sucrose, and organic molecules proteins or enzymes/other suitable named organic molecules;**

With reference to Table 3.1,

(ii) describe the effects of auxin and cytokinin on callus growth. [3]

1. **both cytokinin and auxin needed for cell division/growth / little or no growth if only one present;;**
2. **highest conc. of auxin cf. cytokinin, ratio of 100:1/100X more auxin, only triggers formation of roots growth;;**
3. **lowest conc. of auxin cf. cytokinin, ratio of 4:1/4X more auxin, only triggers formation of shoots;;**
4. **intermediate auxin cf. cytokinin concentration, ratio of 10:1/10X more auxin, triggers formation of more callus/stimulates cell division/growth/mitosis;;**

(iii) Suggest and explain what treatment should be given to the shoots developed from callus in treatment D to turn them into plantlets. [2]

1. **remove from treatment D and implement treatment B/description of treatment B / increase conc of auxin to 50mg dm^{-3} ;;**
2. **stimulates the growth of roots to become a plantlet;;**

Plantlets often do not have fully functional cuticle and this causes them to be inefficient in their use of water. Hence, plantlets might not be able to survive the ex-vivo acclimatisation process if directly transferred from the culture medium to the greenhouse.

(iv) Suggest what steps need to be taken to increase the viability of the plantlets. [2]

1. **Plantlets are removed from the culture vessel;**
2. **washed thoroughly in running tap water to remove traces of agar;**
3. **transferred to plastic cups containing enriched soil covered over by polythene bags for hardening;**
4. **Slowly weaning the plantlets from a high-humidity, low light, warm environment to a normal growth environment for the species;**
5. **After two weeks in enriched soil, the acclimatized plantlets can be transplanted to a potting mixture;**

(d) In many regions of temperate rice production, severe weed infestations have removed land from economic rice production. LLRICE62 is adapted for planting in the temperate rice production regions.

LLRICE62 is a variety of *Oryza sativa* L. rice produced through genetic modification to confer tolerance to the herbicide glufosinate ammonium. Herbicide tolerance is based upon the *bar* gene, a bialaphos resistance gene, isolated from the soil microorganism, *Streptomyces hygroscopicus*. The *bar* gene encodes the production of the enzyme, Phosphinothricin-Acetyl-Transferase (PAT). When sprayed with glufosinate ammonium herbicides, the LLRICE62 plants can continue to grow while surrounding weeds rapidly die.

Based on the above information, suggest

(i) the benefit of cultivating LLRICE62 [1]

1. **Due to less competition from weeds, there is increase in crop yield for the farmers;;**
2. **Allows for removal of weeds that are difficult to control, to make lands available for farming/ allow rice cultivation to return;;**

(ii) the ethical implications of this genetic engineering of LLRICE62. [2]

1. **Violation of natural organisms' intrinsic values;;**
2. **Tampering with nature by mixing genes among species;;**
3. **Introduced genes escaping from a transgenic crop into related weeds through crop-to-weed hybridization, leading to herbicide-resistant weeds.**
(Credit: any other valid points)

[Total: 16]

4 Planning question

You are required to plan, but not carry out, an investigation into the effect of alcohol concentration on the membrane permeability of beetroot tissue. Beetroot is a starchy edible root from the *Beta vulgaris* plant whose cells contain a water-soluble red pigment, betacyanin, in their vacuoles. This pigment cannot pass through membranes unless the membrane is damaged.

Fig. 4.1 shows a spectrophotometer which can be used to measure colour intensity by measuring absorbance of different wavelengths of light as they pass through the sample, as well as the cuvette with sample to be placed into the sample holder. Generally, a sample of higher colour intensity will absorb more light energy at a specific wavelength.

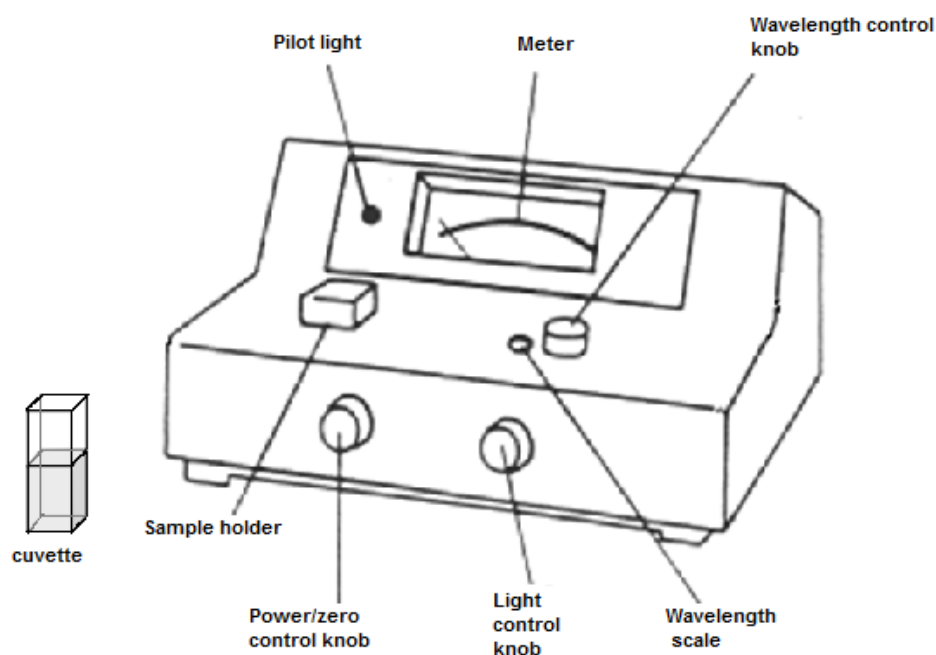


Fig. 4.1

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

Your planning must be based on the assumption that you have been provided with the following equipment and material, which you must use:

- a large piece of beetroot tissue with the skin removed,
- test tubes – 100mm tall with 11mm inside diameter,
- scalpel,
- 10.0% alcohol,
- Spectrophotometer set at 460nm,
- cuvettes,
- distilled water.

You should select from the following apparatus:

- white tile,
- ruler,
- timer e.g. stopwatch or stopclock,
- any normal laboratory glassware e.g. different sized beakers, measuring cylinders or syringes for measuring volume.

[Total: 12]

Mark Scheme for Experiment

- 1. Theoretical consideration or rationale of the plan to justify the practical procedure-prediction**
- 2. Theoretical consideration or rationale of the plan to justify the practical procedure-explanation**
- 3. Identification of variables – independent and dependent variables.**
- 4. Description of method used i.e. using a spectrophotometer, cuvettes, calibration of spectrophotometer using distilled water.**
- 5. Identification of other variables to be kept constant e.g. washing and rinsing cut beetroot cylinders before start of experiment**
- 6. Alcohol concentration as measured by at least 5 different alcohol concentrations (A! range of 1%-10%).**
- 7. Serial dilution shown.**
- 8. Description of how beetroot tissues are to be used in experiment e.g. cutting into cylinders of diameter 5 mm using corkborer, trimming to length of 1.0 cm cylinders OR cut to specific dimensions**
- 9. Description of control.**
- 10. Replicates and at least two repeats.**
- 11. Calculation of mean/average (average calculated in table or in graph).**
- 12. Layout of results table with appropriate headings.**
- 13. Graph with average absorbance/ A.U. (y-axis) against alcohol concentration/ % (x-axis).**
- 14. Safety precautions / risk hazards identified.**
- 15. The correct use of technical and scientific terms.**

[12 marks]

(3) Variables

- The independent variable in this experiment is alcohol concentration while the dependent variable is absorbance.

(5) Other variables to be kept constant (any 1)

- washing and rinsing cut beetroot cylinders before start of experiment.
- calibration of spectrophotometer using distilled water.
- Length (if using corkborer) / dimensions / shape and size of beetroot tissue used (if using scalpel).
- Time the beetroot cylinders were immersed in the alcohol solution.
- Same volume of ethanol used (e.g. 10-15cm³)
- Same cuvette used
- Same beetroot used
- Constant temperature (e.g. 28°C -30°C)
(Any other valid variables)

(1) Prediction of the likely outcome of the experiment

- As concentration of alcohol increases, membrane permeability of beetroot tissue increases, leading to an increase in absorbance value.

(2) Explain the likely outcome of the experiment (b+c)

- Pigment molecule is (a)large/ hydrophilic and cannot pass through (b)hydrophobic core of phospholipid bilayer.
- Increasing alcohol concentration increases (c)damage to/destroy/disrupting the integrity of cell membrane/regular arrangement of the membrane due to (d)denaturation/disruption of tertiary structure of (membrane) proteins/ phospholipids dissolve.
- vacuole membrane/tonoplast AND cell surface membrane (e)lose selective permeability/become more permeable, allowing pigment to diffuse out of the cell.

*(a) to (e) → Pt 15.

(4) Method (any 2)

- Use a spectrophotometer to measure absorbance value.
- Use cuvettes to contain samples.
- Calibration of spectrophotometer with distilled water.

Procedure (includes safety and reliability of results)

1. Prepare 10 cm³ each of 2.0%, 4.0%, 6.0%, 8.0% and 10.0% alcohol solution (Pt 6: 5 conc 0%-10%) and a control of 0.0% alcohol solution (Pt 9: Control) by diluting the 10.0% stock alcohol solution as indicated in the table below. Place them in labelled test tubes.

(7) Dilution table

Test tubes	Concentration of alcohol solution/% (Pt 6: 5 conc 0%-10%)	Volume of 10% alcohol/cm ³	Volume of distilled water/cm ³
A	2.0	2.0	8.0
B	4.0	4.0	6.0
C	6.0	6.0	4.0
D	8.0	8.0	2.0
E	10.0	10.0	0.0
Control (Pt 9)	0.0	0.0	20.0

2. (Pt 8: preparation of beetroot) Cut the beetroot tissue into cylinders of a diameter of 5 mm (less than 11mm) using a cork borer and trim the length to 1.0 cm (less than 100mm) on a white tile using a scalpel OR cut to specific dimensions.
3. (Pt 5: variables kept constant) Wash and rinse beetroot cylinders in distilled water to remove any pigments which may leak out during cutting. There should be no further leakage of colour from the beetroot tissue.
4. (Pt 4: Method) Calibrate the spectrophotometer with cuvette containing distilled water.
5. Place 2 beetroot cylinders (appropriate dimension) into test tube A.
6. (Pt 5: variables kept constant) Immediately start timing for 5 minutes using a stopwatch.
7. (Pt 4: Method) After 5 minutes, remove 1 cm³ of solution from test tube A and transfer to a cuvette to measure absorbance value using a spectrophotometer.

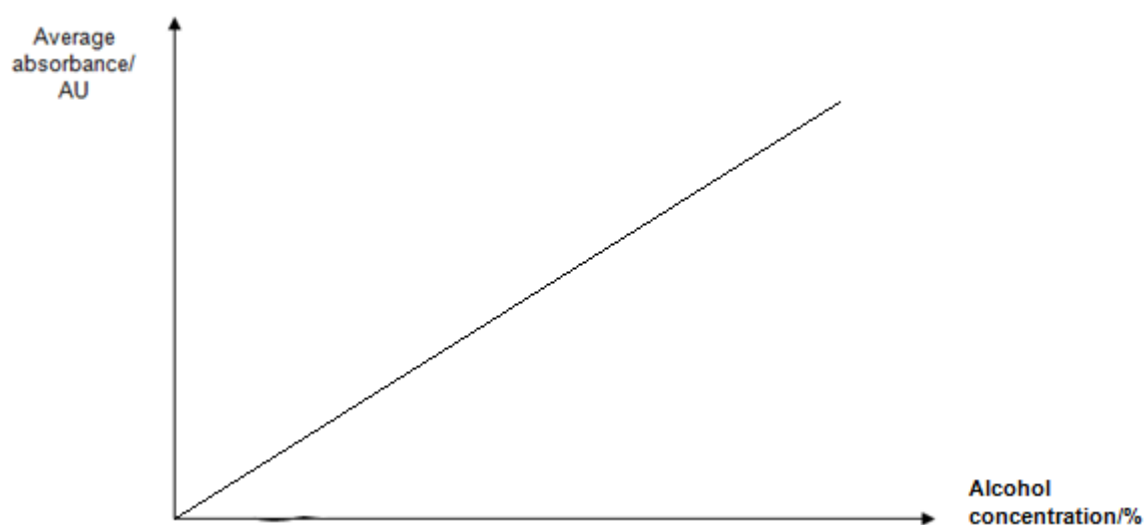
(Pt 10)

8. Repeat step 5-7 to obtain another 2 readings.
9. Repeat step 5-8 for 2%, 4%, 6%, 8%, 10% alcohol solutions.
10. Repeat entire experiment twice.
11. Plot a graph of average absorbance / A.U. against alcohol concentration /%.

(12) Table showing absorbance value at varying alcohol concentrations

Alcohol concentration/%	Absorbance / A.U.			
	Reading 1	Reading 2	Reading 3	Average (Pt 11)
0.0				
2.0				
4.0				
6.0				
8.0				
10.0				

(13) Graph showing average absorbance/ AU against alcohol concentration/ %



(14) Risk Assessment (any 1)

State: Handle sharp objects such as scalpel and cork borer carefully or place them away from main work area after use

Explain: to avoid injuring/cutting oneself.

State: Handle liquid chemicals such as the 10% alcohol with care or wear protective goggles

Explain: to prevent eyes from being in contact with alcohol.

State: Handle fragile objects like test tubes with care. Arrange test tubes neatly in the test tube rack, ensuring that they are placed where they cannot be knocked over or roll off the bench.

Explain: To prevent breakage

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 (a)** Describe the polymerase chain reaction (PCR) and explain the advantages of this procedure. [8]

Stage 1 [2]

1. **Denaturation;**
2. **Hydrogen bonds holding the double-stranded DNA fragment break;**
3. **The double-stranded DNA separate and dissociates into single-stranded DNA;**
4. **By heating to 95°C; (A) 90-100°C**

Stage 2 [2]

5. **Annealing of DNA primers/oligonucleotides;**
6. **DNA primers base pair with complementary sequences;**
7. **At the 3' end of the single stranded DNA;**
8. **By cooling to about 65°C; (A) 30-65°C**

Stage 3 [2]

9. **Primer extension;**
10. **Using the DNA primer, the heat stable polymerase-Taq polymerase synthesizes the rest of the fragment (new strand of DNA);**
11. **Nucleotides are added to the 3' ends on both primers;**
12. **By heating to 72°C; (A) 60-75°C**

Advantages [2]

13. **High sensitivity: PCR can amplify minute amount of DNA, this is especially useful in forensic work where DNA may not be present in high quality;;**
14. **Speed and ease of use: PCR can amplify many copies/very large amounts of DNA in a few hours;;**
15. **Robustness: PCR can permit amplification of specific sequences from material in which the DNA is badly degraded or embedded in a medium from which conventional DNA isolation is problematic;;**
16. **Automation of PCR: use of thermostable Taq polymerase, can withstand high temperatures without being denatured allows for the automation of PCR;;**

(b) Describe, with examples, how a eukaryotic gene can be incorporated into *E.coli*. [6]

- 1. Isolate plasmids;**
- 2. Named example of plasmid-e.g. pBR322/pVector;**
- 3. Named example of human gene-e.g. insulin gene**

Either

- 4. Isolate insulin gene from human DNA;**

Or

- 5. Isolate insulin gene using PCR/reverse transcription using mRNA to form cDNA and add linker DNA;**
- 6. Cut both the human DNA and plasmid with the same restriction enzyme;**
- 7. to get complementary sticky ends;**
- 8. Named example of RE e.g. EcoRI/BamHI/PstI/HindIII;**
- 9. Mix the desired gene and plasmid vector in vitro, allowing the sticky ends to reanneal (in a specific, complementary manner);**
- 10. via complementary base pairing;**
- 11. formation of hydrogen bonds between the complementary bases;**
- 12. DNA ligase is added to the mixture;**
- 13. to seal the nicks between the fragments/join sugar-phosphate backbone together/join adjacent nucleotides together;**
- 14. via the formation of phosphodiester bonds between adjacent nucleotides;**
- 15. so that a recombinant plasmid is formed;**
- 16. Add CaCl_2 to the mixture and subject the mixture to heat shock/electroporation;**
- 17. so that transient pores formed on the bacteria cell membrane;**
- 18. Allowing the recombinant plasmid to enter host *E.coli* cells via transformation;**

- (c) Explain how RFLP analysis facilitates the process of DNA fingerprinting and genomic mapping. [6]

DNA fingerprinting:

1. DNA fingerprinting is a technology that identifies particular individuals using properties of their DNA;
2. Certain loci within human chromosomes contain tandemly repeated sequences (e.g. minisatellites);
3. The number of tandem repeats (at each minisatellite) varies substantially among individuals;
4. and is highly polymorphic;
5. (minisatellites are also referred to as loci with a variable number of tandem repeats (VNTRs));
6. Different number of tandem repeats leads to generation of (different) restriction fragments;
7. that are of different molecular weight and length;
8. giving rise to unique DNA fingerprint;
9. Each individual's combination is unique, having inherited one allele for each locus from each parent;
10. DNA fingerprint of an individual is then analysed / compared against another individual to detect matching band pattern;
- OR
11. Used for forensic analysis for crime investigation / paternity determination / trace evolutionary lineage;

Genomic Mapping:

12. Genomic mapping is a reconstruction of the entire set of chromosomes for a given organism;
13. showing the relative position of every gene / genetic markers on a chromosome;
14. RFLPs greatly increased the number of genetic markers available for mapping the human genome;
Reject: Gene markers / genome markers / marker genes;
15. An RFLP serves as a genetic marker for a particular locus in the genome, (analogous to an allele of a particular gene);
16. RFLPs are used as genetic markers to construct linkage map;
17. The frequency with which two RFLP markers or an RFLP marker and a certain allele for a gene is inherited together is a measure of the closeness of the 2 gene loci on a chromosome and the relative distance between 2 gene loci;;
18. Based on the recombination frequencies / frequency of crossing over between 2 RFLP markers / cross-over values of 2 RFLP markers, the linkage map is constructed;
19. The order of the genetic markers / the relative distances between genetic markers along a chromosome can be determined;